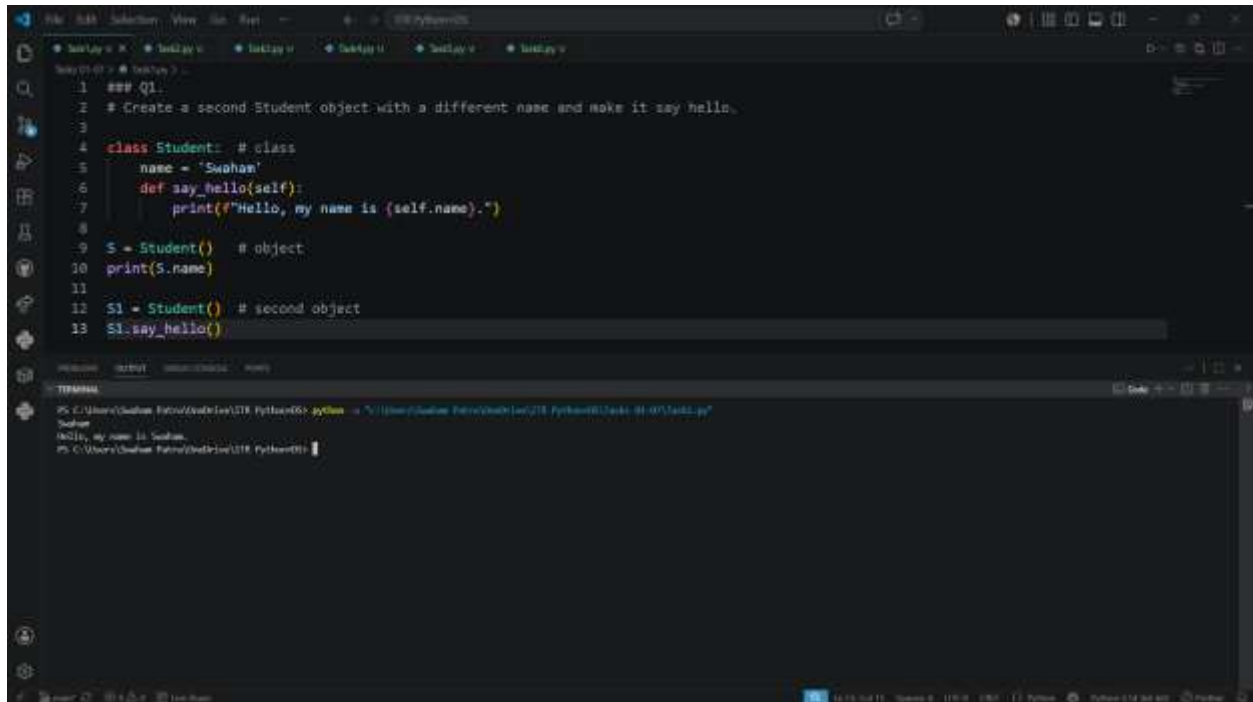


Swaham Pitambar Patra

Task 1:



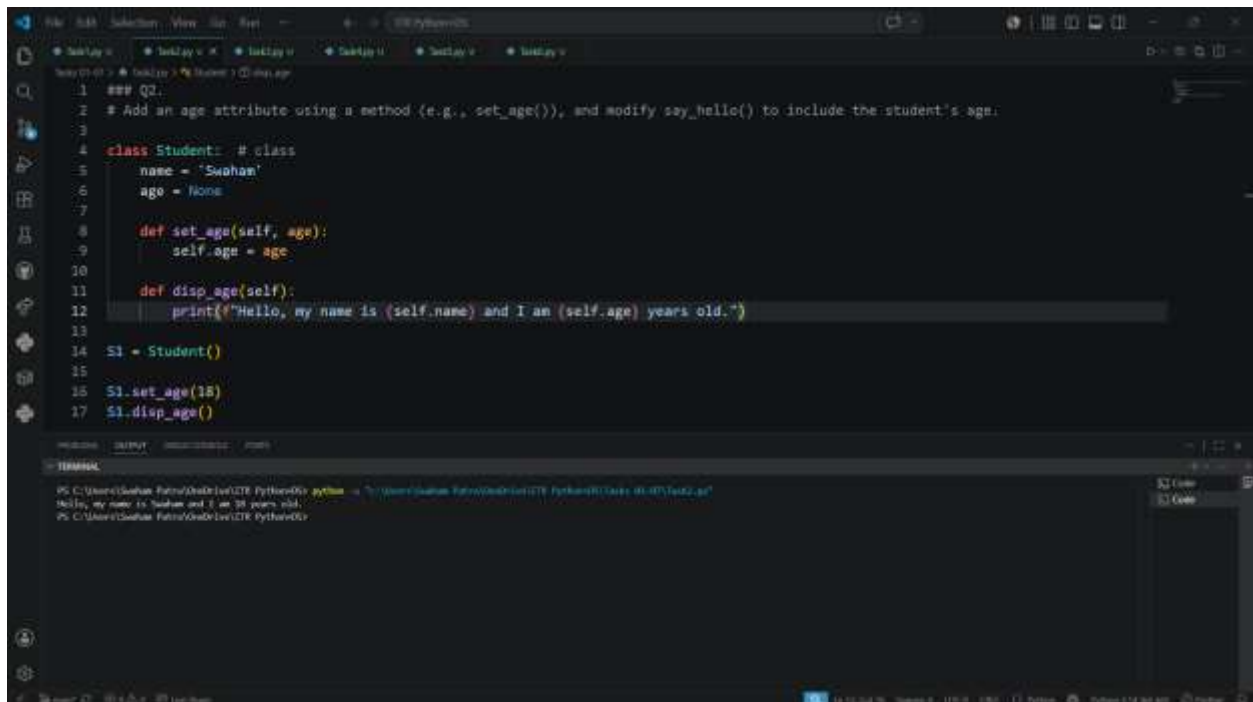
The screenshot shows a Jupyter Notebook interface with a dark theme. The code in the notebook is as follows:

```
1 ### Q1.  
2 # Create a second Student object with a different name and make it say hello.  
3  
4 class Student: # class  
5     name = 'Swahan'  
6     def say_hello(self):  
7         print(f"Hello, my name is {self.name}.")  
8  
9 S = Student() # object  
10 print(S.name)  
11  
12 S1 = Student() # second object  
13 S1.say_hello()
```

The terminal output at the bottom shows the execution results:

```
PS C:\Users\Swaham Patra\OneDrive\Documents> python -i "C:\Users\Swaham Patra\OneDrive\Documents\Python\Task1\Task1.py"  
Swahan  
Hello, my name is Swahan.  
PS C:\Users\Swaham Patra\OneDrive\Documents>
```

Task 2:



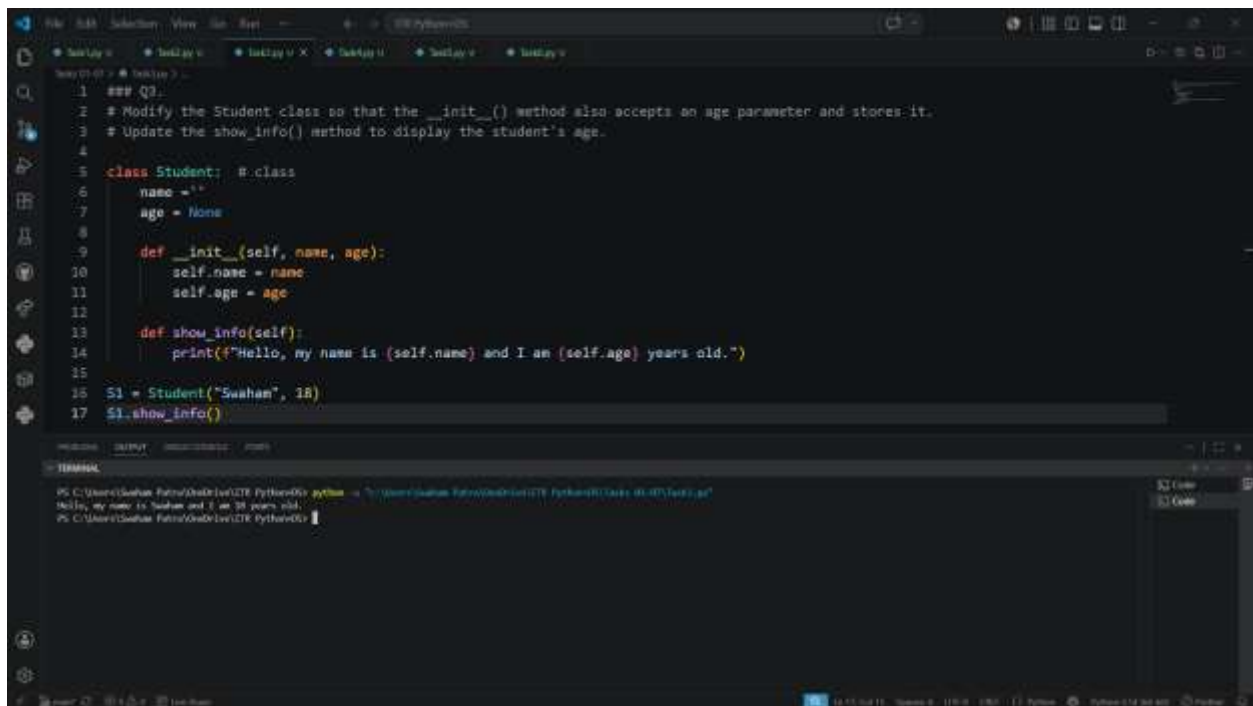
The screenshot shows a Jupyter Notebook interface with a dark theme. The code in the notebook is as follows:

```
1 ### Q2.  
2 # Add an age attribute using a method (e.g., set_age()), and modify say_hello() to include the student's age.  
3  
4 class Student: # class  
5     name = 'Swahan'  
6     age = None  
7  
8     def set_age(self, age):  
9         self.age = age  
10  
11     def disp_age(self):  
12         print(f"Hello, my name is {self.name} and I am {self.age} years old.")  
13  
14 S1 = Student()  
15  
16 S1.set_age(18)  
17 S1.disp_age()
```

The terminal output at the bottom shows the execution results:

```
PS C:\Users\Swaham Patra\OneDrive\Documents> python -i "C:\Users\Swaham Patra\OneDrive\Documents\Python\Task2\Task2.py"  
Hello, my name is Swahan and I am 18 years old.  
PS C:\Users\Swaham Patra\OneDrive\Documents>
```

Task 3:



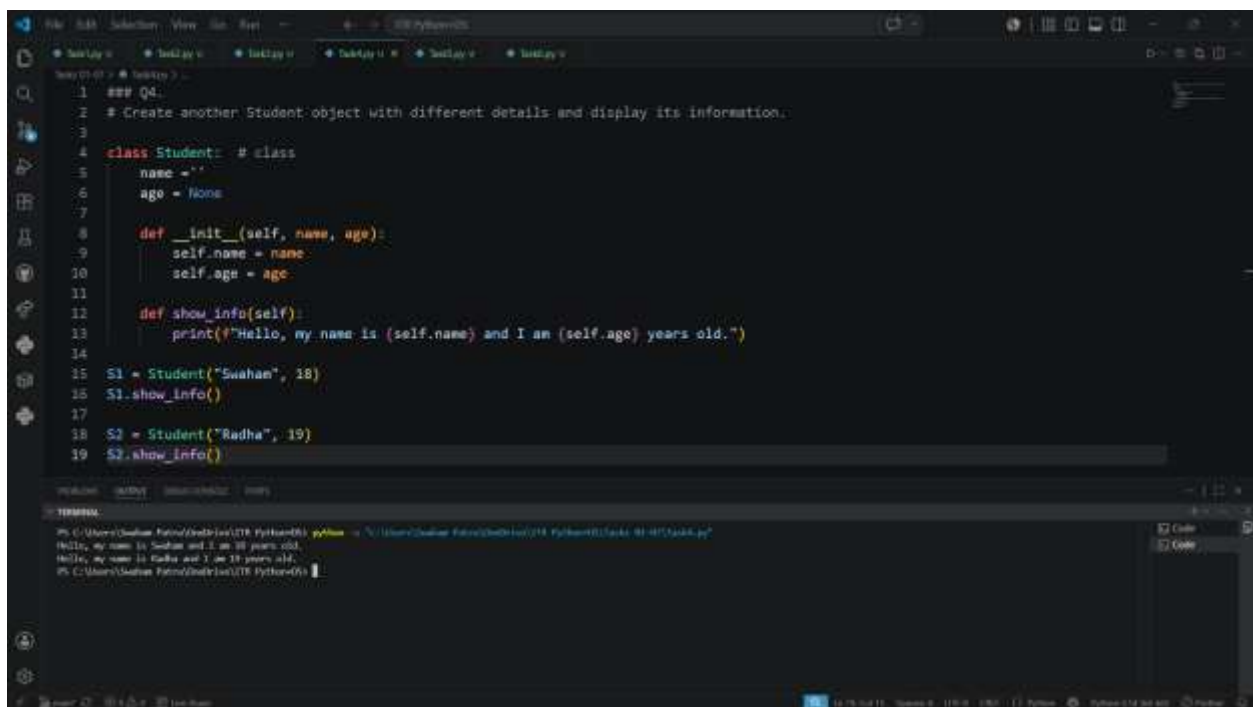
The screenshot shows a Python IDE with a dark theme. The editor window displays a Python script for Task 3. The code defines a `Student` class with an `__init__` method that takes `name` and `age` as arguments and a `show_info` method that prints the student's name and age. An instance `S1` of the `Student` class is created with the name "Swahan" and age 18, and its `show_info` method is called. The terminal window at the bottom shows the output of the script, which is "Hello, my name is Swahan and I am 18 years old."

```
1 ### Q3.
2 # Modify the Student class so that the __init__() method also accepts an age parameter and stores it.
3 # Update the show_info() method to display the student's age.
4
5 class Student: # class
6     name = ''
7     age = None
8
9     def __init__(self, name, age):
10         self.name = name
11         self.age = age
12
13     def show_info(self):
14         print(f"Hello, my name is {self.name} and I am {self.age} years old.")
15
16 S1 = Student("Swahan", 18)
17 S1.show_info()
```

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```
PS C:\Users\Swahan Patil\OneDrive\Python> python -u C:\Users\Swahan Patil\OneDrive\Python\Tasks\03-Q3\Task3.py
Hello, my name is Swahan and I am 18 years old.
PS C:\Users\Swahan Patil\OneDrive\Python>
```

Task 4:



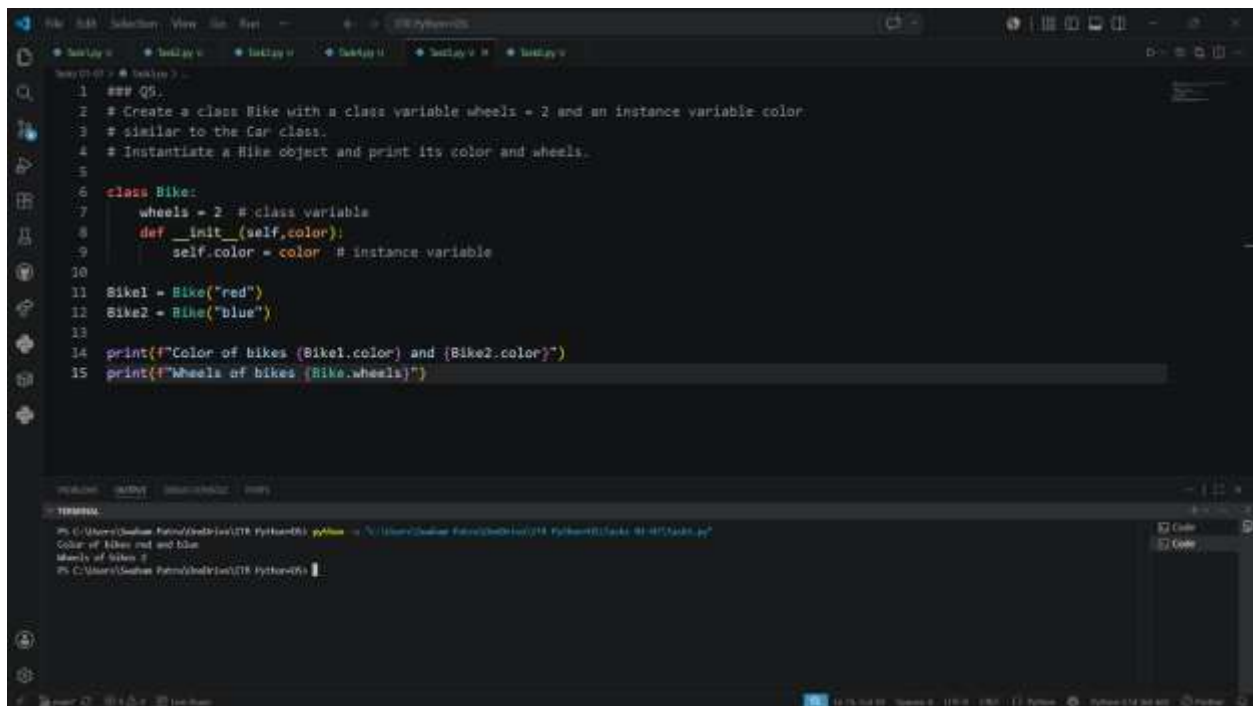
The screenshot shows a Python IDE with a dark theme. The editor window displays a Python script for Task 4. The code defines a `Student` class with an `__init__` method that takes `name` and `age` as arguments and a `show_info` method that prints the student's name and age. Two instances of the `Student` class are created: `S1` with the name "Swahan" and age 18, and `S2` with the name "Radha" and age 19. The `show_info` method is called for both instances. The terminal window at the bottom shows the output of the script, which is "Hello, my name is Swahan and I am 18 years old." followed by "Hello, my name is Radha and I am 19 years old."

```
1 ### Q4.
2 # Create another Student object with different details and display its information.
3
4 class Student: # class
5     name = ''
6     age = None
7
8     def __init__(self, name, age):
9         self.name = name
10        self.age = age
11
12     def show_info(self):
13         print(f"Hello, my name is {self.name} and I am {self.age} years old.")
14
15 S1 = Student("Swahan", 18)
16 S1.show_info()
17
18 S2 = Student("Radha", 19)
19 S2.show_info()
```

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```
PS C:\Users\Swahan Patil\OneDrive\Python> python -u C:\Users\Swahan Patil\OneDrive\Python\Tasks\04-Q4\Task4.py
Hello, my name is Swahan and I am 18 years old.
Hello, my name is Radha and I am 19 years old.
PS C:\Users\Swahan Patil\OneDrive\Python>
```

Task 5:

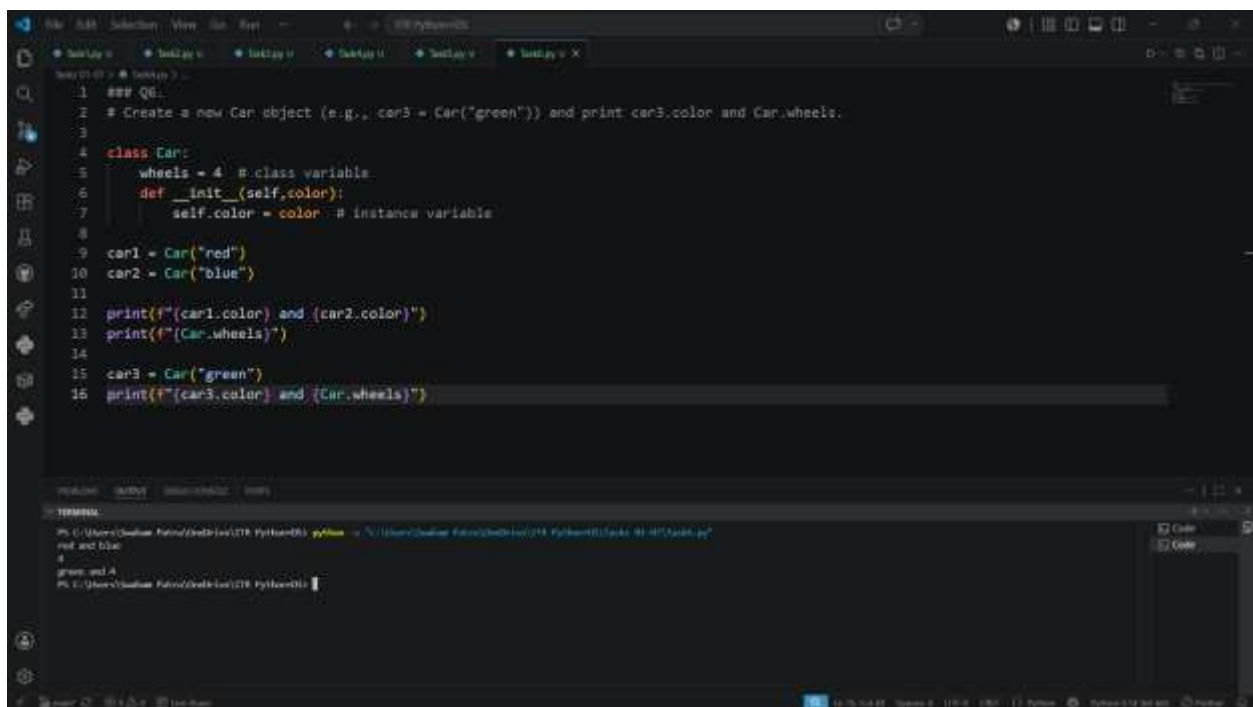


```
1 ### Q5.  
2 # Create a class Bike with a class variable wheels = 2 and an instance variable color.  
3 # similar to the Car class.  
4 # Instantiate a Bike object and print its color and wheels.  
5  
6 class Bike:  
7     wheels = 2 # class variable  
8     def __init__(self,color):  
9         self.color = color # instance variable  
10  
11 Bike1 = Bike("red")  
12 Bike2 = Bike("blue")  
13  
14 print(f"Color of bikes {Bike1.color} and {Bike2.color}")  
15 print(f"Wheels of bikes {Bike.wheels}")
```

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```
PS C:\Users\Gadwin Parni\OneDrive\OneDrive\Python> python -u %cd%\Gadwin Parni\OneDrive\OneDrive\Python\Task 5\Task5.py  
Color of bikes red and blue  
Wheels of bikes 2  
PS C:\Users\Gadwin Parni\OneDrive\OneDrive\Python>
```

Task 6:



```
1 ### Q6.  
2 # Create a new Car object (e.g., car3 = Car("green")) and print car3.color and Car.wheels.  
3  
4 class Car:  
5     wheels = 4 # class variable  
6     def __init__(self,color):  
7         self.color = color # instance variable  
8  
9 car1 = Car("red")  
10 car2 = Car("blue")  
11  
12 print(f"{car1.color} and {car2.color}")  
13 print(f"{Car.wheels}")  
14  
15 car3 = Car("green")  
16 print(f"{car3.color} and {Car.wheels}")
```

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```
PS C:\Users\Gadwin Parni\OneDrive\OneDrive\Python> python -u %cd%\Gadwin Parni\OneDrive\OneDrive\Python\Task 6\Task6.py  
red and blue  
4  
green and 4  
PS C:\Users\Gadwin Parni\OneDrive\OneDrive\Python>
```